

GIRRAWEE HIGH SCHOOL

MATHEMATICS

YEAR 12 HSC

Task 2, 2012

Time Allowed: 90 minutes

Name: _____

Instructions:

Examiner: C. McMillan

- Attempt all questions
- Circle the best response for the questions in Part A
- Start each question in Part B on a new page
- All necessary working must be shown
- Marks may be deducted for careless or badly arranged work

PART A (5 marks)

For questions 1-5 circle the best response from the following:

Question 1: The probability that a traffic light will turn green as a vehicle approaches it is estimated to be $\frac{5}{12}$. A taxi goes through 192 intersections where there are traffic lights. How many of these would be expected to turn green as the taxi approaches:

- A) 80 B) 112 C) 110 D) 192

Question 2: All x values for which the curve $f(x) = x^2 - 4x + 1$ is increasing are:

- A) $x \leq 2$ B) $x \geq 2$ C) $x < 2$ D) $x > 2$

Question 3: Given $h = 5t^3 - 2t^2 + t + 5$, the value of $\frac{d^2h}{dt^2}$ when $t = 1$ is:

- A) 9 B) 12 C) 26 D) 27

Question 4: The primitive function of $(2x-1)^2$ is:

- A) $\frac{(2x-1)^3}{6} + C$ B) $\frac{(2x-1)^3}{3} + C$ C) $\frac{(2x-1)^3}{2} + C$ D) $(2x-1)^3 + C$

Question 5: The value of $\int_{-1}^0 x^3 dx$ is:

- A) $\frac{1}{4}$ B) $-\frac{1}{4}$ C) 1 D) -1

PART B

Question 1 (11 marks)

(a) A book has 124 pages. If the book is opened at any page at random, find the probability of the page number being:

- i) either 80 or 90 (1)
- ii) a multiple of 10 (2)
- iii) an odd number (1)
- iv) less than 100 (1)

(b) In a group of 75 students, altogether 54 do History and 31 do Geography.
If I select one student at random, find the probability that the student will do:

- i) only Geography (2)
- ii) both History and Geography (2)

(c) In Yahtzee, 5 dice are rolled. Find the probability of rolling five 6's (2)

Question 2 (11 marks)

(a) Find any stationary points on the curve $f(x) = 2x^3 - 15x^2 + 24x - 7$ and determine their nature. (5)

(b) Find all values of x for which the curve is concave up given that $y = x^3 - x^2 + x + 9$. (3)

(c) If $y = x^2$ show that $2y \left(\frac{d^2y}{dx^2} \right) = \left(\frac{dy}{dx} \right)^2$. (3)

Question 3 (9 marks)

(a) Find the second derivative of $y = \sqrt{3x-1}$. (3)

(b) Sketch $f(x) = 2x^3 + 3x^2 - 36x + 5$ for $-3 \leq x \leq 3$, showing any stationary points and points of inflexion. Find the maximum and minimum values of the function in $-3 \leq x \leq 3$. (6)

Question 4 (11 marks)

- (a) The council wanted to make a rectangular swimming area at the beach using a straight cliff on one side and a length of 300m of shark proof netting for the other three sides. What are the dimensions of the rectangle that encloses the greatest area. (5)

- (b) A poster consists of a photograph bordered by a 5cm margin. The area of the poster is to be 400cm^2 .

- (i) Show that the area of the photograph is given by

$$A = 500 - 10x - \frac{4000}{x} \quad (3)$$

- (ii) Find the maximum area possible for the photograph. (3)

Question 5 (22 marks)

- (a) Find each indefinite integral:

(i) $\int (y - 3) dy$ (1)

(ii) $\int \frac{dx}{x^3}$ (2)

(iii) $\int \sqrt{x} \left(1 + \frac{1}{\sqrt{x}} \right) dx$ (3)

(iv) $\int (4 + 3x)^4 dx$ (1)

(v) $\int \sqrt{(5x + 2)^5} dx$ (3)

- (b) Evaluate:

(i) $\int_0^2 \left(\frac{4x^3 + x^2 + 5x}{x} \right) dx$ (4)

(ii) $\int_1^4 \sqrt{x} dx$ (3)

(iii) $\int_4^5 (5 - x)^6 dx$ (2)

(iv) $\int_0^1 \frac{dx}{(3x - 2)^4}$ (3)

Question 6 (12 marks)

(a) If $y = ax^3 - 12x^2 + 3x - 5$ has a point of inflexion at $x = 2$, evaluate a . (3)

(b) Given that the gradient of the tangent to a curve is given by $\frac{dy}{dx} = 2 - 6x$ and the curve passes through $(-2, 3)$, find the equation of the curve. (3)

(c) A function has a tangent parallel to the line $4x - y - 2 = 0$ at the point $(0, -2)$ and $f''(x) = 12x^2 - 6x + 4$. Find the equation of the function. (6)

END OF PAPER.

PART A

- 1) A
- 2) D
- 3) C
- 4) A
- 5) B

PART B

Question 1

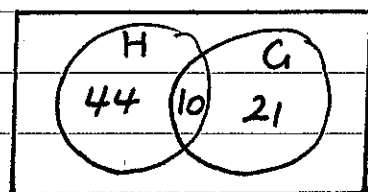
a) $P(\text{either 80 or 90}) = \frac{1}{62}$

ii) $P(X < 10) = \frac{12}{124} = \frac{3}{31}$

iii) $P(\text{odd number}) = \frac{1}{2}$

iv) $P(< 100) = \frac{99}{124}$

b)



i) $P(\text{only Geography}) = \frac{21}{75}$
 $= \frac{7}{25}$

ii) $P(\text{both History and Geography})$
 $= \frac{2}{15}$

c) $P(\text{five 6's}) = \left(\frac{1}{6}\right)^5 = \frac{1}{7776}$

Question 2

a) $f'(x) = 6x^2 - 30x + 24$

$6(x^2 - 5x + 4) = 0$

$\therefore (x-4)(x-1) = 0$

$\therefore x = 4, 1$

stationary pts.

When $x = 4$, $y = 2(4)^3 - 15(4)^2 + 24(4) - 7$
 $= -23$

When $x = 1$, $y = 2(1)^3 - 15(1)^2 + 24(1) - 7$
 $= 4$

$\therefore$ stat. pts at $(4, -23), (1, 4)$

$f''(x) = 12x - 30$

When $x = 4$, $f''(x) > 0 \therefore$ min. turning pt

$x = 1$, $f''(x) < 0 \therefore$ max. turning pt

$\therefore (4, -23)$ is a minimum turning pt and $(1, 4)$ is a maximum turning pt.

b) $y' = 3x^2 - 2x + 1$

$y'' = 6x - 2$

$y'' > 0$

$6x - 2 > 0$

$6x > 2 \therefore x > \frac{1}{3}$

Question 2 cont.

$$c) y = x^2$$

$$\frac{dy}{dx} = 2x$$

$$\frac{d^2y}{dx^2} = 2$$

$$2y \left(\frac{d^2y}{dx^2} \right) = \left(\frac{dy}{dx} \right)^2$$

$$\text{LHS} = 2y \left(\frac{d^2y}{dx^2} \right)$$

$$= 2(x^2)(2)$$

$$= 4x^2$$

$$\text{RHS} = \left(\frac{dy}{dx} \right)^2$$

$$= (2x)^2$$

$$= 4x^2$$

$$= \text{LHS}$$

$$\therefore 2y \left(\frac{d^2y}{dx^2} \right) = \left(\frac{dy}{dx} \right)^2$$

Question 3

$$a) y = \sqrt{3x-1}$$

$$= (3x-1)^{\frac{1}{2}}$$

$$\frac{dy}{dx} = \frac{1}{2} (3x-1)^{-\frac{1}{2}} \times 3$$

$$= \frac{3}{2} (3x-1)^{-\frac{1}{2}}$$

$$= \frac{3}{2\sqrt{3x-1}}$$

$$\frac{d^2y}{dx^2} = -\frac{3}{4} (3x-1)^{-\frac{3}{2}} \times 3$$

$$= -\frac{9}{4} (3x-1)^{-\frac{3}{2}}$$

$$= -\frac{9}{4\sqrt{(3x-1)^3}}$$

Question 3

a) $y' = 3x^2 - 12x + 5$

$y'' = 6x - 12$

pt of inflexion when $y'' = 0$

$6x - 12 = 0$

$6x = 12$

$x = 2$

possible pt of inflexion

Test:

x	1.9	2	2.1
y''	-0.6	0	0.6

When $x = 2$

$y = 2^3 - 6(2)^2 + 5(2) + 9$

$= 8 - 24 + 10 + 9$

$= 3$

∴ A point of inflexion is at $(2, 3)$.

b) Stationary pts.

$f'(x) = 6x^2 + 6x - 36$

$6(x^2 + x - 6) = 0$

$\therefore (x+3)(x-2) = 0$

$\therefore x = -3, 2$

When $x = -3$

$f(x) = 2(-3)^3 + 3(-3)^2 - 36(-3) + 5$

$= 86$

When $x = 2$

$f(x) = 2(2)^3 + 3(2)^2 - 36(2) + 5$

$= -39$

When $x = 3$

$f(x) = 2(3)^3 + 3(3)^2 - 36(3) + 5 = -22$

$f''(x) = 12x + 6$

When $x = -3$, $f''(x) < 0$

maximum

When $x = 2$, $f''(x) > 0 \therefore$ minimum

$\therefore (-3, 86)$ maximum turning pt
and $(2, -39)$ minimum turning pt

$f''(x) = 0$

$12x + 6 = 0$

$\therefore x = -\frac{1}{2}$ possible pt of inflexion

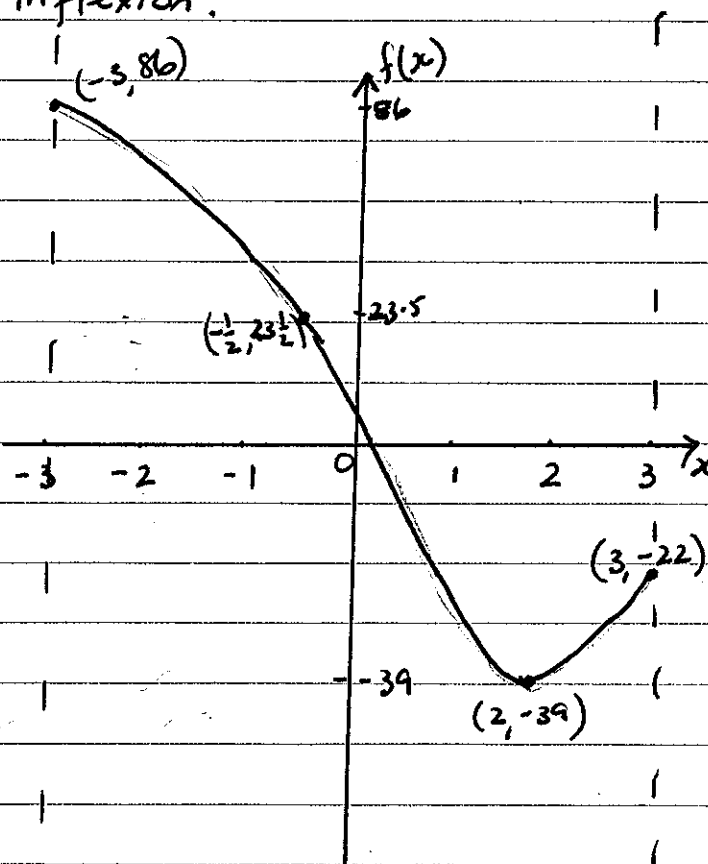
Test:

x	-0.51	-0.5	-0.49
f''(x)	0.12	0	0.12

When $x = -0.5$

$f(x) = 23.5$

$\therefore (-0.5, 23.5)$ is a point of inflexion.

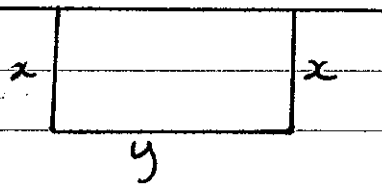


$\therefore$ minimum value = -39

maximum value = 86.

Question 4.

a)



let:

width be x

length be y .

$$\text{Perimeter: } 300 = 2x + y$$

$$\therefore y = 300 - 2x$$

$$\text{Area} = x(300 - 2x)$$

$$= 300x - 2x^2$$

$$\frac{dA}{dx} = 300 - 4x$$

When $\frac{dA}{dx} = 0$

$$300 - 4x = 0$$

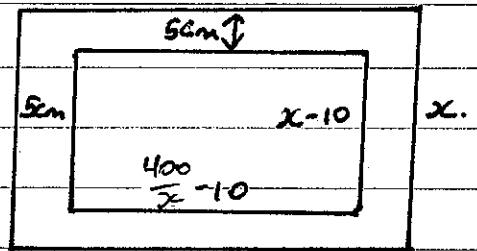
$$4x = 300$$

$$x = 75$$

$$\frac{d^2A}{dx^2} = -4 \text{ always max.}$$

$\therefore$ The width is 75m
and the length is 150m.

b)



Poster

$$A = 400\text{cm}^2$$

$$\therefore xy = 400$$

$$\therefore y = \frac{400}{x}$$

$$\text{Photo Area} = \left(\frac{400}{x} - 10\right)(x - 10)$$

$$= 400 - \frac{4000}{x} - 10x + 100$$

$$= 500 - \frac{4000}{x} - 10x$$

$$\therefore A = 500 - 10x - \frac{4000}{x}$$

$$\text{i) } \frac{dA}{dx} = -10 + \frac{4000}{x^2}$$

$$-10 + \frac{4000}{x^2} = 0$$

$$\frac{4000}{x^2} = 10$$

$$10x^2 = 4000$$

$$x^2 = 400$$

$$\therefore x = 20$$

$$\frac{d^2A}{dx^2} = -\frac{4000}{x^3}$$

$$\text{When } x = 20, \frac{d^2A}{dx^2} < 0$$

$\therefore$ maximum.

$$\text{Area Photo} = 500 - 200 - 200 = 100\text{cm}^2$$

Question 5

$$\text{ai) } \int (y-3) dy$$

$$= \frac{y^2}{2} - 3y + c$$

$$\text{ii) } \int \frac{dx}{x^3}$$

$$= \int x^{-3} dx$$

$$= -\frac{x^{-2}}{2}$$

$$= -\frac{1}{2x^2} + c$$

$$\text{iii) } \int \sqrt{x} \left(1 + \frac{1}{\sqrt{x}}\right) dx$$

$$= \int \sqrt{x} + 1 dx$$

$$= \int x^{\frac{1}{2}} + 1 dx$$

$$= \frac{2}{3} x^{\frac{3}{2}} + x$$

$$= \frac{2}{3} \sqrt{x^3} + x$$

$$= \frac{2x\sqrt{x} + x}{3} + c$$

$$\text{iv) } \int (4+3x)^4 dx$$

$$= \frac{(4+3x)^5}{15} + c$$

$$\text{v) } \int \sqrt{(5x+2)^5} dx$$

$$= \int (5x+2)^{\frac{5}{2}} dx$$

$$= \frac{(5x+2)^{\frac{7}{2}}}{5 \times \frac{7}{2}}$$

$$= \frac{2 \sqrt{(5x+2)^7}}{35} + c$$

$$\text{bi) } \int_0^2 \left(\frac{4x^3 + x^2 + 5x}{x} \right) dx$$

$$= \int_0^2 (4x^2 + x + 5) dx$$

$$= \left[\frac{4x^3}{3} + \frac{x^2}{2} + 5x \right]_0^2$$

$$= \left(\frac{32}{3} + 2 + 10 \right) - 0$$

$$= \frac{68}{3}$$

$$\text{ii) } \int_1^4 \sqrt{x} dx$$

$$= \left[\frac{2}{3} x^{\frac{3}{2}} \right]_1^4$$

$$= \left(\frac{2}{3} (4)^{\frac{3}{2}} \right) - \left(\frac{2}{3} (1)^{\frac{3}{2}} \right)$$

$$= \frac{14}{3}$$

Question 5 cont.

$$\text{iii) } \int_4^5 (5-x)^6 dx$$

$$= \left[-\frac{(5-x)^7}{7} \right]_4^5$$

$$= +\frac{1}{7}$$

$$\text{iv) } \int_0^1 \frac{dx}{(3x-2)^4} dx$$

$$= \int_0^1 (3x-2)^{-4} dx$$

$$= \left[-\frac{(3x-2)^{-3}}{9} \right]_0^1$$

$$= \left[-\frac{1}{9(3x-2)^3} \right]_0^1$$

$$= \left(-\frac{1}{9} \right) - \left(-\frac{1}{72} \right)$$

$$= -\frac{9}{72} = -\frac{1}{8}$$

Question 6

$$\text{a) } y' = 3ax^2 - 24x + 3$$

$$y'' = 6ax - 24$$

$$y'' = 0 \text{ at } x = 2$$

$$0 = 12a - 24$$

$$12a = 24$$

$$\therefore a = 2$$

$$\text{b) } \frac{dy}{dx} = 2 - 6x$$

$$y = 2x - 3x^2 + c$$

passes through $(-2, 3)$

$$3 = 2(-2) - 3(-2)^2 + c$$

$$3 = -4 - 12 + c$$

$$\therefore c = 19$$

$$\therefore y = 19 + 2x - 3x^2$$

$$\text{c) } 4x - y - 2 = 0$$

$$\therefore y = 4x - 2$$

$$m_{\text{tangent}} = 4 \text{ at } (0, -2)$$

$$f''(x) = 12x^2 - 6x + 4$$

$$f'(x) = 4x^3 - 3x^2 + 4x + c$$

$$f'(x) = 4 \text{ at } (0, -2)$$

$$4 = \frac{12(0)^3}{3} - 3(0)^2 + 4(0) + c$$

$$\therefore c = 4$$

$$\therefore f'(x) = 4x^3 - 3x^2 + 4x + 4$$

$$f(x) = x^4 - x^3 + 2x^2 + 4x + c$$

$$\text{When } x = 0 \quad f(x) = -2.$$

$$-2 = 0^4 - 0^3 + 2(0)^2 + 4(0) + c$$

$$\therefore c = -2.$$

$$\therefore f(x) = x^4 - x^3 + 2x^2 + 4x - 2.$$